

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
|
from datetime import datetime

pd.set_option("display.max_columns", None)
pd.set_option("display.max_rows", 100)

print("Libraries imported successfully!")
```

Libraries imported successfully!

```
] df = pd.read_csv("ecommerce_customer_data_custom_ratios.csv")  
  
print("Dataset loaded successfully!")
```

Dataset loaded successfully!

```
[8]: print("Number of Rows:", df.shape[0])
      print("Number of Columns:", df.shape[1])

      print("\nColumn Names:")
      print(df.columns.tolist())
```



```
Number of Rows: 250000
```

```
Number of Columns: 13
```

```
Column Names:
```

```
['Customer ID', 'Purchase Date', 'Product Category', 'Product Price', 'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age', 'Returns', 'Customer Name', 'Age', 'Gender', 'Churn']
```

```
[9]: df.head()
```



[9]:

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	46251	2020-09-08 09:38:32	Electronics	12	3	740	Credit Card	37	0.0	Christine Hernandez	37	Male	0
1	46251	2022-03-05 12:56:35	Home	468	4	2739	PayPal	37	0.0	Christine Hernandez	37	Male	0
2	46251	2022-05-23 18:18:01	Home	288	2	3196	PayPal	37	0.0	Christine Hernandez	37	Male	0
3	46251	2020-11-12 13:13:29	Clothing	196	1	3509	PayPal	37	0.0	Christine Hernandez	37	Male	0
4	13593	2020-11-27 17:55:11	Home	449	1	3452	Credit Card	49	0.0	James Grant	49	Female	1

[]:

df.tail()													
	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
249995	33308	2023-08-10 13:39:06	Clothing	279	2	2187	PayPal	55	1.0	Michelle Flores	55	Male	1
249996	48835	2021-11-23 01:30:42	Home	27	1	3615	Credit Card	42	1.0	Jeremy Rush	42	Female	1
249997	21019	2020-07-02 14:04:48	Home	17	5	2466	Cash	41	0.0	Tina Craig	41	Male	0
249998	49234	2020-12-30 02:02:40	Books	398	2	3668	Crypto	34	0.0	Jennifer Cooper	34	Female	1
249999	16971	2021-03-13 16:28:35	Electronics	425	4	2370	Cash	36	1.0	Justin Lawson	36	Female	1

df.sample(10)													
	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
117175	21108	2023-06-24 08:07:44	Home	233	3	3590	PayPal	56	NaN	Rita Williams	56	Female	1
208433	33961	2022-10-24 05:10:19	Home	303	1	3610	Cash	48	1.0	Christopher Grant DVM	48	Female	0
158359	3624	2022-06-27 22:07:20	Home	312	4	2979	Credit Card	60	1.0	Timothy Hicks	60	Female	0
237185	12276	2022-08-11 01:36:12	Clothing	169	4	643	PayPal	43	NaN	Benjamin Sanchez	43	Female	0
86519	23893	2022-11-20 13:18:10	Home	231	5	4049	Cash	62	1.0	Christopher Munoz	62	Male	1
55569	32142	2021-01-07 11:38:10	Clothing	219	5	1827	PayPal	67	NaN	Mr. Jonathan Sawyer	67	Male	0
14237	12563	2022-12-28 04:07:50	Electronics	73	5	4498	Cash	54	NaN	Danielle Alexander	54	Male	1
121874	26760	2020-12-21 09:10:57	Books	475	5	277	Credit Card	31	NaN	Toni Medina	31	Male	0

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 250000 entries, 0 to 249999
```

```
Data columns (total 13 columns):
```

#	Column	Non-Null Count	Dtype
0	Customer ID	250000 non-null	int64
1	Purchase Date	250000 non-null	object
2	Product Category	250000 non-null	object
3	Product Price	250000 non-null	int64
4	Quantity	250000 non-null	int64
5	Total Purchase Amount	250000 non-null	int64
6	Payment Method	250000 non-null	object
7	Customer Age	250000 non-null	int64
8	Returns	202404 non-null	float64
9	Customer Name	250000 non-null	object
10	Age	250000 non-null	int64
11	Gender	250000 non-null	object
12	Churn	250000 non-null	int64

```
dtypes: float64(1), int64(7), object(5)
```

```
memory usage: 24.8+ MB
```

```

]: missing = df.isnull().sum()

missing_percentage = (
    df.isnull().sum() / len(df) * 100
).round(2)

missing_table = pd.DataFrame({
    "Missing Values": missing,
    "Missing %": missing_percentage
})

missing_table

```

```

]:

```

	Missing Values	Missing %
Customer ID	0	0.00
Purchase Date	0	0.00
Product Category	0	0.00
Product Price	0	0.00
Quantity	0	0.00
Total Purchase Amount	0	0.00

Payment Method	0	0.00
Customer Age	0	0.00
Returns	47596	19.04
Customer Name	0	0.00
Age	0	0.00
Gender	0	0.00
Churn	0	0.00

```
: duplicate_count = df.duplicated().sum()  
  
print("Duplicate rows:", duplicate_count)
```

Duplicate rows: 0

```
: unique_customers = df["Customer ID"].nunique()
```

```
print("Unique Customers:", unique_customers)
```

```
Unique Customers: 49673
```

```
[16]: df["Purchase Date"] = pd.to_datetime(df["Purchase Date"])

print("Minimum Date:", df["Purchase Date"].min())
print("Maximum Date:", df["Purchase Date"].max())
```

Minimum Date: 2020-01-01 00:15:00

Maximum Date: 2023-09-15 12:24:08

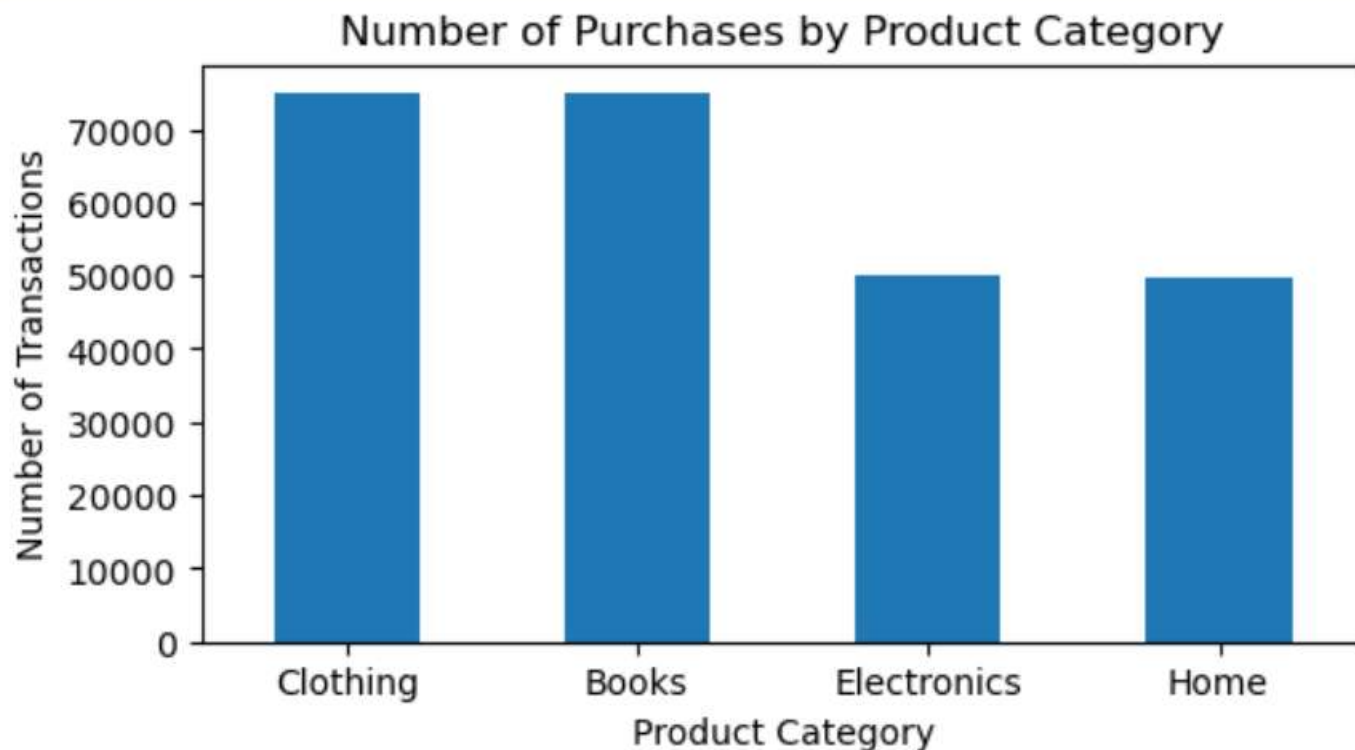
```
21]: category_counts = df["Product Category"].value_counts()  
  
print(category_counts)
```

```
Product Category  
Clothing          75052  
Books             74912  
Electronics       50185  
Home              49851  
Name: count, dtype: int64
```

```
23]: plt.figure(figsize=(6,3))

category_counts.plot(kind="bar")

plt.title("Number of Purchases by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Number of Transactions")
plt.xticks(rotation=0)
|
plt.show()
```



```
payment_counts = df["Payment Method"].value_counts()
```

```
payment_counts
```

```
Payment Method
```

```
Credit Card    100486
```

```
PayPal         74837
```

```
Cash           49894
```

```
Crypto         24783
```

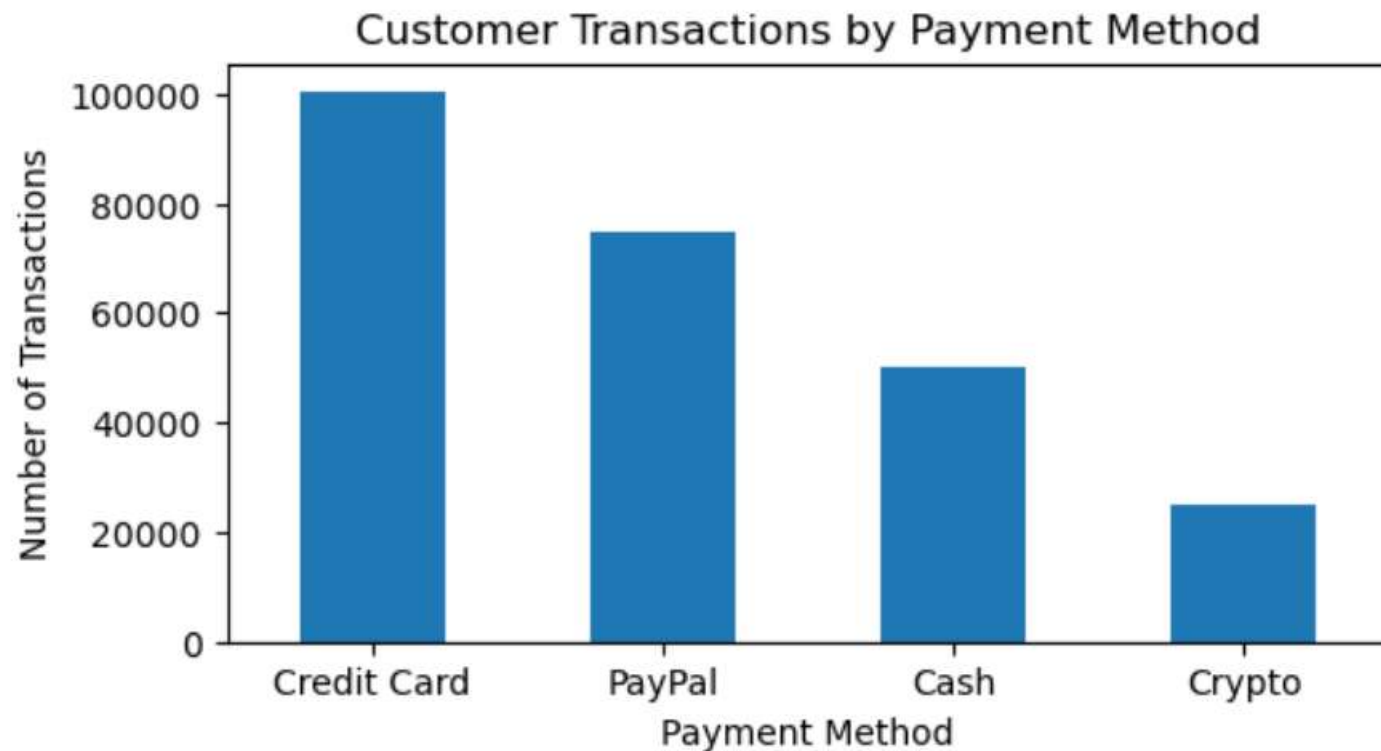
```
Name: count, dtype: int64
```

```
[26]: plt.figure(figsize=(6,3))

payment_counts.plot(kind="bar")

plt.title("Customer Transactions by Payment Method")
plt.xlabel("Payment Method")
plt.ylabel("Number of Transactions")
plt.xticks(rotation=0)

plt.show()
```



```
gender_counts = df["Gender"].value_counts()  
|  
gender_counts
```

Gender

Female 125560

Male 124440

Name: count, dtype: int64

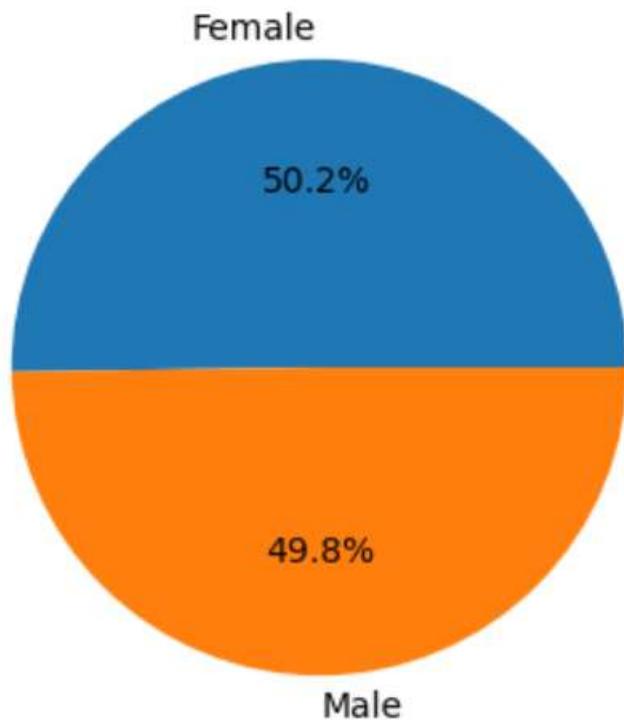
```
[9]: plt.figure(figsize=(7,4))

gender_counts.plot(kind="pie", autopct="%1.1f%%")

plt.title("Customer Gender Distribution")
plt.ylabel("")

plt.show()
```

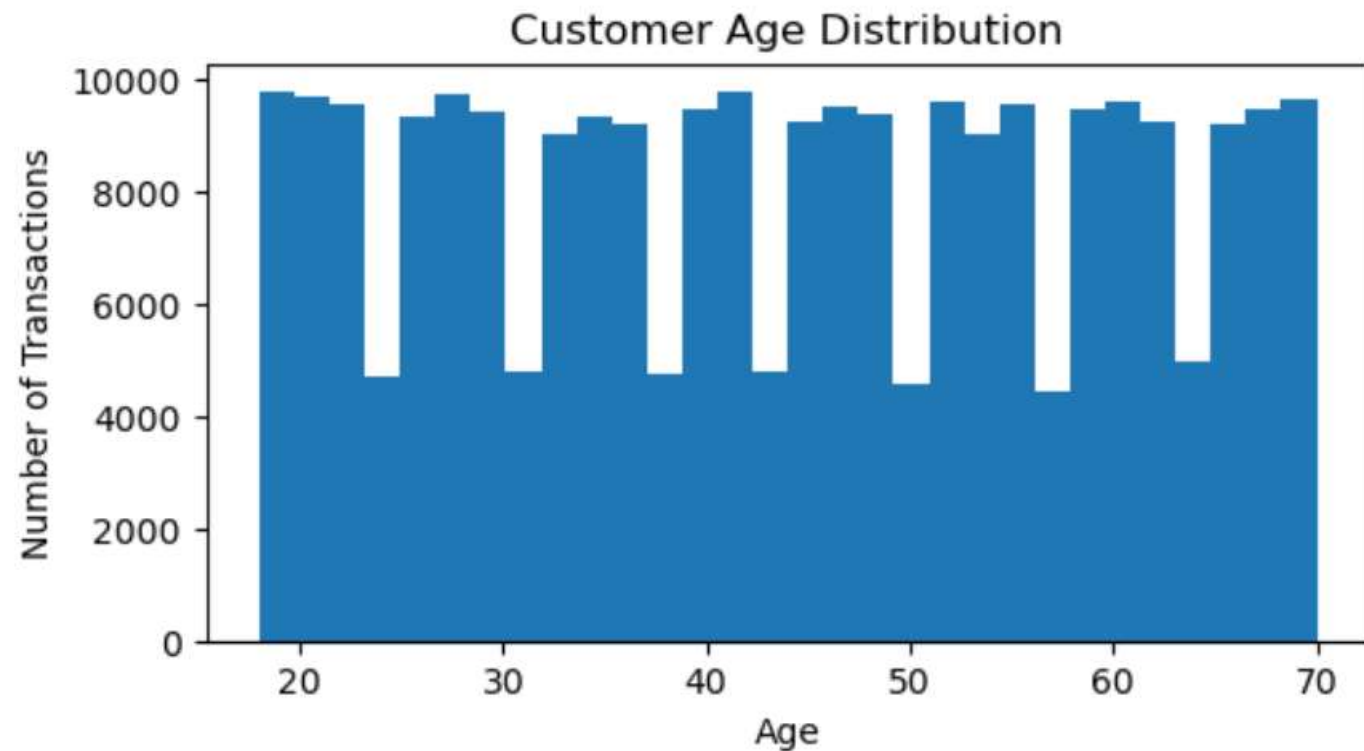
Customer Gender Distribution



```
plt.figure(figsize=(6,3))

plt.hist(df["Customer Age"], bins=30)

plt.title("Customer Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Transactions")
|
plt.show()
```



```
34]: df["Total Purchase Amount"].describe()
```

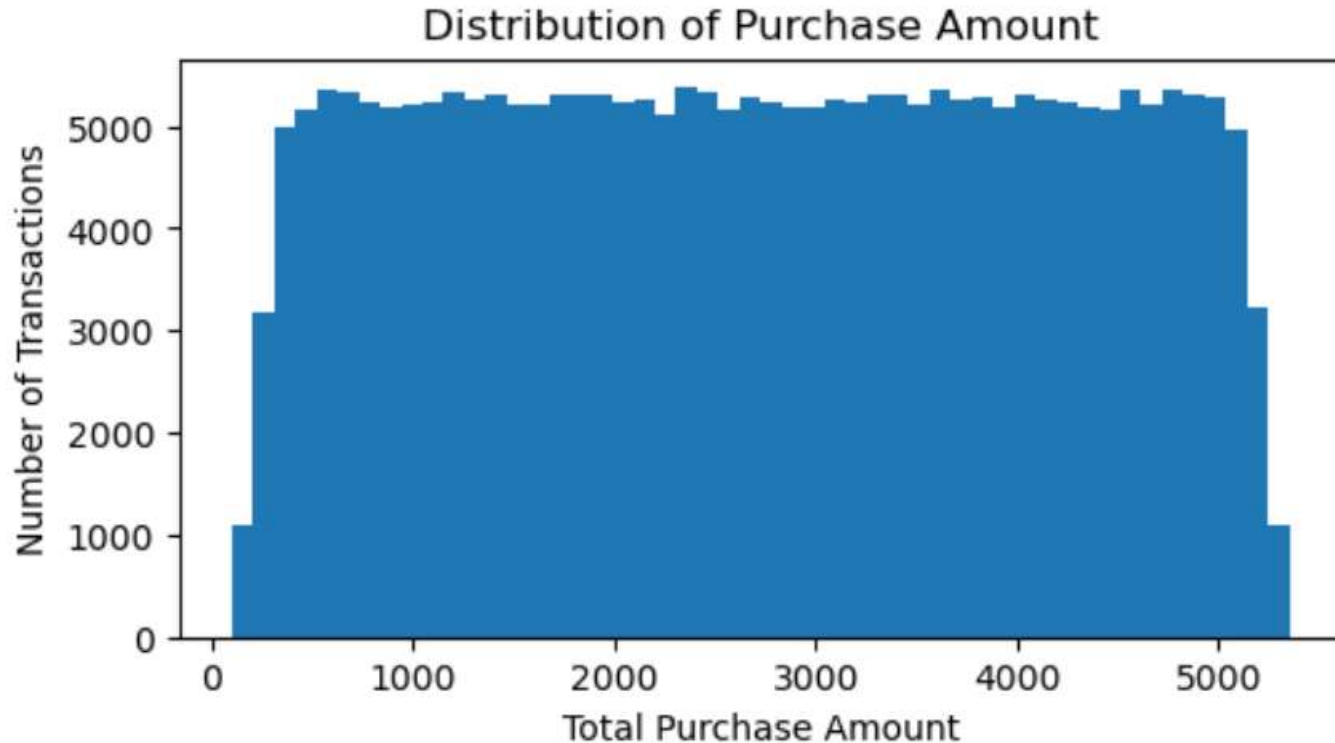
```
34]: count      250000.000000  
      mean       2725.370732  
      std        1442.933565  
      min         100.000000  
      25%        1477.000000  
      50%        2724.000000  
      75%        3974.000000  
      max         5350.000000  
      Name: Total Purchase Amount, dtype: float64
```

```
plt.figure(figsize=(6,3))

plt.hist(df["Total Purchase Amount"], bins=50)

plt.title("Distribution of Purchase Amount")
plt.xlabel("Total Purchase Amount")
plt.ylabel("Number of Transactions")

plt.show()
```



```
7]: df["Quantity"].describe()
```

```
7]: count      250000.000000  
    mean         2.998896  
    std          1.414694  
    min          1.000000  
    25%          2.000000  
    50%          3.000000  
    75%          4.000000  
    max          5.000000  
    Name: Quantity, dtype: float64
```

```
8]: df["Quantity"].value_counts().sort_index()
```

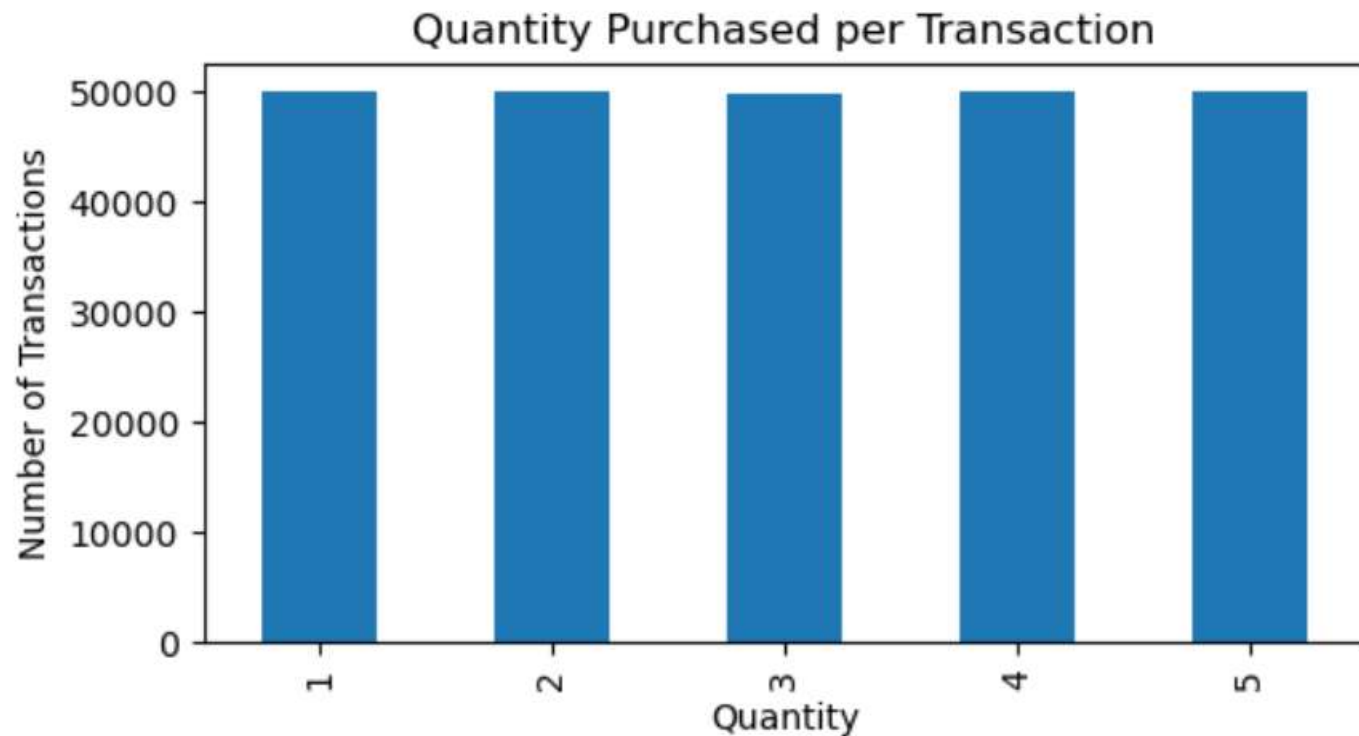
```
8]: Quantity  
    1      50108  
    2      50021  
    3      49872  
    4      50037  
    5      49962  
    Name: count, dtype: int64
```

```
[40]: plt.figure(figsize=(6,3))

df["Quantity"].value_counts().sort_index().plot(kind="bar")

plt.title("Quantity Purchased per Transaction")
plt.xlabel("Quantity")
plt.ylabel("Number of Transactions")

plt.show()
```



```
[41]: churn_counts = df["Churn"].value_counts()

print(churn_counts)
```

```
Churn
0      200126
1       49874
Name: count, dtype: int64
```

```
[42]: churn_rate = df["Churn"].mean() * 100

print(f"Churn Rate: {churn_rate:.2f}%")
```

```
Churn Rate: 19.95%
```

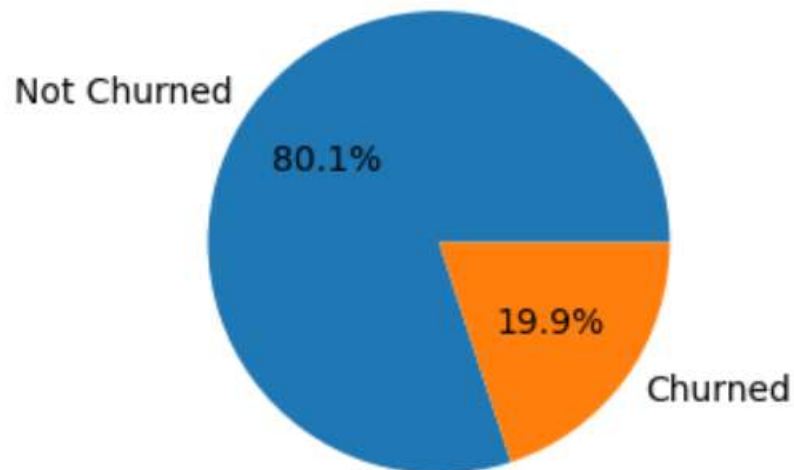
```
[43]: plt.figure(figsize=(6,3))

churn_counts.plot(
    kind="pie",
    autopct="%1.1f%%",
    labels=["Not Churned", "Churned"]
)

plt.title("Customer Churn Distribution")
plt.ylabel("")

plt.show()
```

Customer Churn Distribution



```
[44]: print(df["Returns"].value_counts(dropna=False))
```

Returns

0.0 101635

1.0 100769

NaN 47596

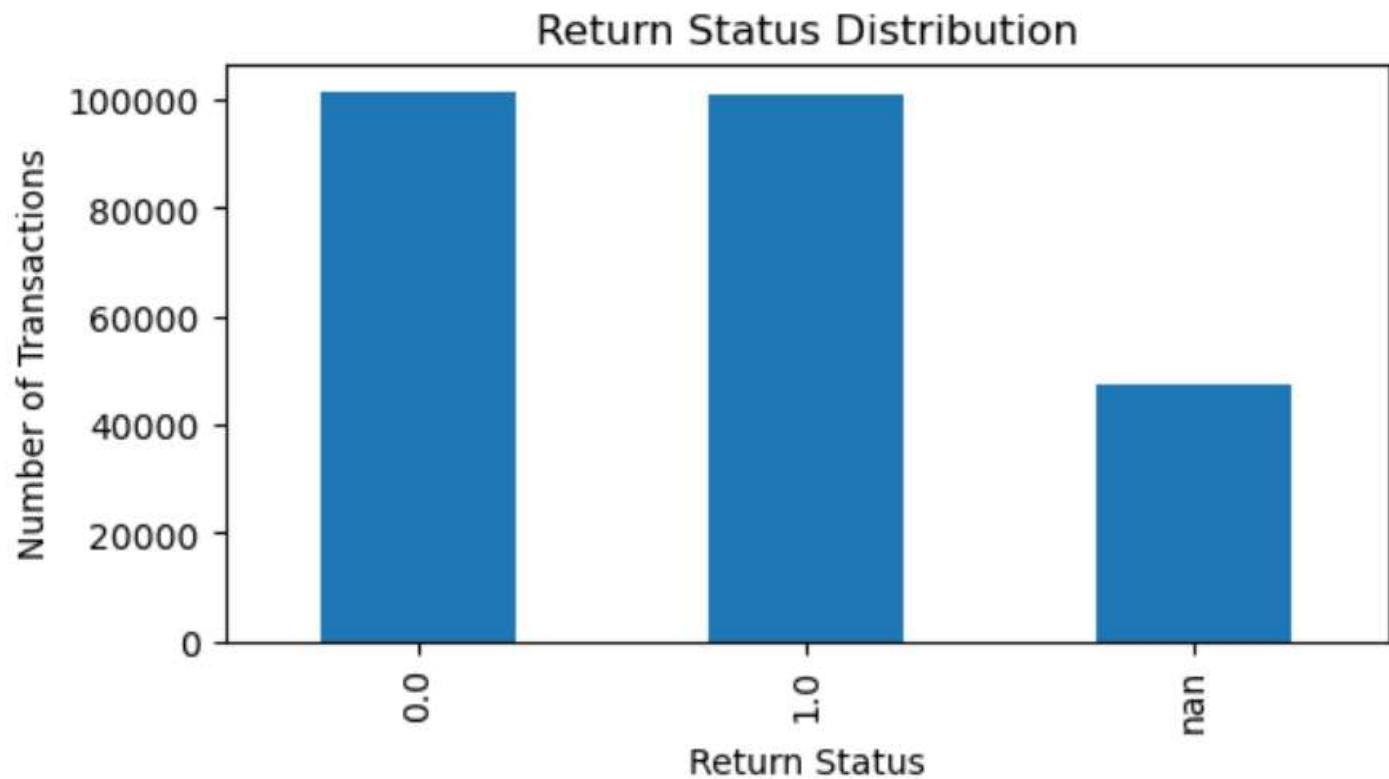
Name: count, dtype: int64

```
[46]: plt.figure(figsize=(6,3))

df["Returns"].value_counts(dropna=False).plot(kind="bar")

plt.title("Return Status Distribution")
plt.xlabel("Return Status")
plt.ylabel("Number of Transactions")

plt.show()
```



```
[49]: numeric_columns = [  
        "Product Price",  
        "Quantity",  
        "Total Purchase Amount",  
        "Customer Age",  
        "Churn"  
    ]  
  
    correlation = df[numeric_columns].corr()  
  
    plt.figure(figsize=(6,3))  
  
    sns.heatmap(  
        correlation,  
        annot=True,  
        fmt=".2f"  
    )  
  
    plt.title("Correlation Matrix")  
  
    plt.show()
```

Correlation Matrix

